

CSS Grid Layout Module

The **CSS Grid Layout Module** is a revolutionary tool in the realm of web development. It introduces a **grid-based layout system**, facilitated by rows and columns, that simplifies the design process of any webpage.

This system eliminates the need for floats and positioning, making it a game-changer for developers.

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Document</title>
    <style>
      .container {
        width: 100%;
        border: 4px solid red;
        display: grid;
        /* grid-template-columns: 200px; */
        /* grid-template-columns: 200px 200px; */
        /* grid-template-columns: 200px 200px 100px; */
        grid-template-columns: 100px 200px auto;
        /* grid-template-rows: 50px; */
        /* grid-template-rows: 50px 50px; */
        grid-template-rows: 50px auto;
        grid-column-gap: 5px;
        grid-row-gap: 5px;
      }
      #item1 {
        background-color: rgb(240, 126, 126);
      }
      #item2 {
        background-color: rgb(34, 53, 220);
      }
      #item3 {
        background-color: rgb(207, 20, 76);
      }
      #item4 {
        background-color: rgb(239, 174, 78);
      }
      #item5 {
        background-color: rgb(30, 192, 225);
      }
    </style>
  </head>
  <body>
    <div class="container">
      <div id="item1">Item 1</div>
      <div id="item2">Item 2</div>
      <div id="item3">Item 3</div>
      <div id="item4">Item 4</div>
      <div id="item5">Item 5</div>
    </div>
  </body>
</html>
```

```
#item6 {
  background-color: rgb(207, 213, 25);
}
</style>
</head>
<body>
  <div class="container">
    <div class="item" id="item1">1st</div>
    <div class="item" id="item2">2nd</div>
    <div class="item" id="item3">3rd</div>
    <div class="item" id="item4">4th</div>
    <div class="item" id="item5">5th</div>
    <div class="item" id="item6">6th</div>
  </div>
</body>
</html>
```

Display: grid

```
<style>
  .container {
    border: 4px solid red;
    display: grid;
    grid-template-columns: auto auto auto;
    padding: 5px;
  }
  .item {
    background-color: rgba(255, 255, 255, 0.8);
    border: 1px solid black;
    padding: 20px;
    font-size: 30px;
    text-align: center;
  }
</style>
```

Display:inline-grid

```
<style>
  .container {
    border: 4px solid red;
    display: inline-grid;
    grid-template-columns: auto auto auto;
    padding: 5px;
  }
  .item {
    background-color: rgba(255, 255, 255, 0.8);
    border: 1px solid black;
    padding: 20px;
    font-size: 30px;
    text-align: center;
  }
</style>
```

We can control the gaps in the grid system using these items

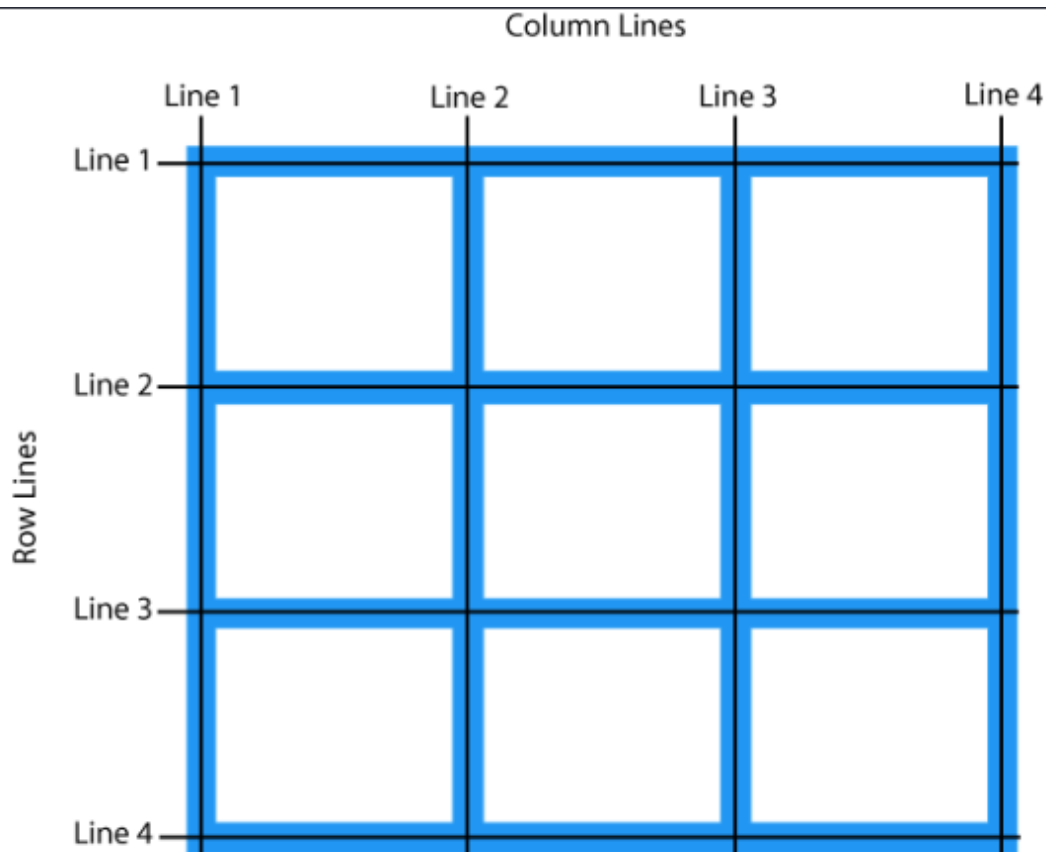
- grid-column-gap
- grid-row-gap
- grid-gap

```
<style>
  .container {
    border: 4px solid red;
    display: grid;
    grid-template-columns: auto auto auto;
    padding: 5px;
    grid-column-gap: 50px;
    grid-row-gap: 50px;
  }
  .item {
    background-color: rgba(255, 255, 255, 0.8);
    border: 1px solid black;
    padding: 20px;
    font-size: 30px;
    text-align: center;
  }
</style>
```

Grid Lines

The lines between columns are called *column lines*.

The lines between rows are called *row lines*.



Refer to line numbers when placing a grid item in a grid container:

Grid col start And Grid col end:

```
<style>
.container {
  border: 4px solid red;
  background-color: gray;
  display: grid;
  grid-template-columns: auto auto auto;
  padding: 5px;
  grid-column-gap: 50px;
  grid-row-gap: 50px;
}
```

```
}  
.item {  
  background-color: rgba(255, 255, 255, 0.8);  
  border: 1px solid black;  
  padding: 20px;  
  font-size: 30px;  
  text-align: center;  
}  
#item1 {  
  grid-column-start: 1;  
  grid-column-end: 3;  
}  
#item3 {  
  grid-row-start: 2;  
  grid-row-end: 5;  
}  
</style>
```